

P-BLOCK MASTER STATEMENT ASSIGNMENT

GROUP 16 + GROUP 17 •

PART I — GROUP 16

Q1. Occurrence and natural abundance [Count the correct statements.]

- (1) Oxygen, sulphur, selenium, tellurium and polonium constitute Group 16.
- (2) Group 16 is also known as the chalcogen family; the name is derived from a Greek word for brass and points to the association of sulphur and its congeners with copper.
- (3) Oxygen is the most abundant of all the elements on earth and forms about 46.6% by mass of the earth's crust.
- (4) Dry air contains about 20.946% oxygen by volume.
- (5) The abundance of sulphur in the earth's crust is only about 0.03-0.1%.
- (6) Gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, is an important naturally occurring sulphate.
- (7) Epsom salt, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, is a sulphur-containing mineral.
- (8) Baryte has the formula BaSO_4 .
- (9) Galena, zinc blende and copper pyrites are represented by PbS , ZnS and CuFeS_2 respectively.
- (10) Traces of sulphur occur as hydrogen sulphide in volcanoes.
- (11) Eggs, proteins, garlic, onion, mustard, hair and wool contain sulphur.
- (12) Selenium and tellurium occur as metal selenides and tellurides in sulphide ores.
- (13) Polonium occurs in nature as a decay product of thorium and uranium minerals.
- (14) Sulphur is more abundant than oxygen in the earth's crust.
- (15) Most copper minerals contain either oxygen or sulphur and frequently contain other members of the group as well.

Answer: _____

Q2. Electronic configuration, size and ionisation enthalpy [Count the correct statements.]

- (1) The general valence-shell electronic configuration of Group 16 is ns^2np^4 .
- (2) The Group 16 elements have six electrons in their outermost shell.
- (3) Atomic radii generally increase from oxygen to polonium.
- (4) Ionic radii of E^{2-} ions generally increase down the group.
- (5) The oxygen atom is exceptionally small compared with the heavier members.
- (6) First ionisation enthalpy generally decreases down the group.
- (7) The decrease in ionisation enthalpy down the group is mainly related to increasing atomic size.
- (8) Group 16 elements have lower first ionisation enthalpies than the corresponding Group 15 elements.
- (9) The extra stability of the half-filled p^3 configuration helps explain the higher ionisation enthalpy of Group 15.
- (10) The first ionisation enthalpy increases regularly from O to Po.
- (11) Oxygen has the largest atomic radius in Group 16.
- (12) Group 16 elements have higher first ionisation enthalpies than Group 15 elements in the same periods.
- (13) The general valence-shell electronic configuration is ns^2np^5 .

(14) Increasing number of electron shells is responsible for the increase in size down the group.

(15) The ionic radius of O^{2-} is larger than that of Te^{2-} .

Answer: _____

Q3. EXCLUSIVE ORDER QUESTION - periodic properties [Count the correct statements.]

- (1) Atomic radius: $\text{O} < \text{S} < \text{Se} < \text{Te} < \text{Po}$.
- (2) E^{2-} ionic radius: $\text{O}^{2-} < \text{S}^{2-} < \text{Se}^{2-} < \text{Te}^{2-} < \text{Po}^{2-}$.
- (3) First ionisation enthalpy: $\text{O} > \text{S} > \text{Se} > \text{Te} > \text{Po}$.
- (4) Electronegativity: $\text{O} > \text{S} > \text{Se} > \text{Te} > \text{Po}$.
- (5) Electron gain enthalpy is most negative for sulphur among O, S, Se, Te and Po.
- (6) From S to Po, electron gain enthalpy becomes progressively less negative.
- (7) Metallic character: $\text{O} < \text{S} < \text{Se} < \text{Te} < \text{Po}$.
- (8) Stability of the -2 oxidation state decreases down the group.
- (9) Stability of the +6 oxidation state decreases down the group.
- (10) Stability of the +4 oxidation state increases down the group.
- (11) Acidic character of H_2E : $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$.
- (12) Thermal stability of H_2E decreases on moving down the group.
- (13) Reducing character of H_2S , H_2Se and H_2Te increases down the group.
- (14) H-E bond dissociation enthalpy: $\text{H-O} > \text{H-S} > \text{H-Se} > \text{H-Te}$.
- (15) Electron gain enthalpy becomes steadily more negative from O to Po.

Answer: _____

Q4. Atomic and physical data - Data mastery I [Count the correct statements.]

- (1) Atomic numbers of O, S, Se, Te and Po are 8, 16, 34, 52 and 84 respectively.
- (2) Approximate atomic masses for O, S, Se, Te and Po are 16.00, 32.06, 78.96, 127.60 and 210.00 g mol⁻¹.
- (3) Covalent radii for O, S, Se, Te and Po are 66, 104, 117, 137 and 146 pm respectively.
- (4) E^{2-} ionic radii for O, S, Se, Te and Po are 140, 184, 198, 221 and about 230 pm respectively.
- (5) Electron gain enthalpy values for O and S are about -141 and -200 kJ mol⁻¹ respectively.
- (6) Electron gain enthalpy values for Se, Te and Po are about -195, -190 and -174 kJ mol⁻¹ respectively.
- (7) First ionisation enthalpies for O and S are about 1314 and 1000 kJ mol⁻¹.
- (8) First ionisation enthalpies for Se, Te and Po are about 941, 869 and 813 kJ mol⁻¹.
- (9) The covalent radius of oxygen is 104 pm.
- (10) The electron gain enthalpy of oxygen is more negative than that of sulphur.
- (11) The first ionisation enthalpy of polonium exceeds that of oxygen.
- (12) The electronegativities are approximately O 3.50, S 2.58, Se 2.55, Te 2.01 and Po 1.76.
- (13) Tellurium is as more electronegative than selenium.
- (14) The numerical data show a much larger size jump from O to S than between several successive heavier members.
- (15) The electron gain enthalpy of polonium is as -274 kJ mol⁻¹.

Answer: _____

Q5. Physical properties, allotropy and atomicity [Count the correct statements.]

- (1) Oxygen and sulphur are non-metals.
- (2) Selenium and tellurium are described as metalloids.
- (3) Polonium is metallic.
- (4) Polonium is radioactive and short lived.
- (5) All Group 16 elements exhibit allotropy.
- (6) Melting and boiling points generally increase with atomic number down the group.

- (7) A large difference between oxygen and sulphur melting/boiling points is related to their different atomicities.
- (8) Elemental oxygen exists as O_2 whereas common sulphur exists as S_8 .
- (9) Oxygen exists as O_8 in its ordinary elemental state.
- (10) Sulphur is a diatomic molecule under ordinary conditions.
- (11) Polonium is described as a non-metal.
- (12) Selenium and tellurium are described as typical metals.
- (13) The melting point of oxygen is about 55 K.
- (14) The boiling point of oxygen is about 90 K.
- (15) The boiling point of sulphur is lower than that of oxygen.

Answer: _____

Q6. Oxidation states and inert-pair trend [Count the correct statements.]

- (1) The stability of the -2 oxidation state decreases down Group 16.
- (2) Polonium hardly shows the -2 oxidation state.
- (3) Oxygen generally shows the -2 oxidation state in its compounds.
- (4) Oxygen has positive oxidation states in compounds with fluorine.
- (5) The oxidation state of oxygen in OF_2 is +2.
- (6) The oxidation state of oxygen in O_2F_2 is +1 per oxygen atom.
- (7) Sulphur, selenium and tellurium commonly show +4 in compounds with oxygen.
- (8) Sulphur, selenium and tellurium can show +6, especially in compounds with fluorine.
- (9) The +6 oxidation state becomes less stable down the group.
- (10) The +4 oxidation state becomes more stable down the group because of the inert pair effect.
- (11) Bonding in the +4 and +6 oxidation states is primarily covalent.
- (12) The +6 oxidation state becomes more stable from S to Po.
- (13) Oxygen commonly expands its valence shell to show +6.
- (14) The +4 oxidation state becomes less stable down the group.
- (15) Polonium most commonly exhibits a stable -2 state according to the trend.

Answer: _____

Q7. Anomalous behaviour of oxygen [Count the correct statements.]

- (1) The anomalous behaviour of oxygen is related to its very small size.
- (2) Its high electronegativity contributes to the anomalous behaviour.
- (3) Strong hydrogen bonding in H_2O is an important consequence of oxygen's small size and high electronegativity.
- (4) Comparable strong hydrogen bonding is not present in H_2S .
- (5) Oxygen has no valence-shell d orbitals.
- (6) Absence of d orbitals limits the covalency of oxygen.
- (7) In practice, the covalency of oxygen rarely exceeds two.
- (8) Heavier Group 16 elements can expand their valence shells so that covalency can exceed four.
- (9) Oxygen readily uses 2d orbitals for expansion of its valence shell.
- (10) H_2S has stronger hydrogen bonding than H_2O .
- (11) The anomalous behaviour of oxygen is due mainly to its very large size.
- (12) Oxygen is the second most electronegative element after fluorine.
- (13) Oxygen has the lowest electronegativity of Group 16.

- (14) The compactness of oxygen also explains why its electron gain enthalpy is less negative than that of sulphur.
- (15) The electron gain enthalpy of oxygen is more negative than that of sulphur because oxygen is smaller.

Answer: _____

Q8. Hydrides H_2E - trends and chemistry [Count the correct statements.]

- (1) All Group 16 elements form hydrides of the type H_2E .
- (2) Acidic character increases from H_2O to H_2Te .
- (3) The increase in acidity down the group is related to decreasing H-E bond dissociation enthalpy.
- (4) Thermal stability decreases from H_2O down the group.
- (5) All hydrides except water possess reducing character.
- (6) Reducing character increases from H_2S to H_2Te .
- (7) H_2Te is more acidic than H_2S .
- (8) H_2Te is a stronger reducing agent than H_2S .
- (9) H_2O is the strongest reducing hydride of the group.
- (10) Thermal stability increases from H_2O to H_2Te .
- (11) H_2S is more acidic than H_2Te .
- (12) The H-E bond length increases from H_2O to H_2Te .
- (13) The H-E-H bond angle decreases toward 90° on moving down the series.
- (14) The anomalously high boiling point of H_2O is associated with hydrogen bonding.
- (15) H_2O has the weakest H-E bond in the series H_2O , H_2S , H_2Se , H_2Te .

Answer: _____

Q9. Hydride data - exact numerical trends [Count the correct statements.]

- (1) The melting points of H_2O , H_2S , H_2Se and H_2Te are about 273, 188, 208 and 222 K.
- (2) The boiling points of H_2O , H_2S , H_2Se and H_2Te are about 373, 213, 232 and 269 K.
- (3) The H-E distances for H_2O , H_2S , H_2Se and H_2Te are about 96, 134, 146 and 169 pm.
- (4) The H-E-H angles are about 104° , 92° , 91° and 90° respectively.
- (5) The H-E bond dissociation enthalpies are about 463, 347, 276 and 238 kJ mol^{-1} .
- (6) The first dissociation constants increase strongly from H_2O to H_2Te .
- (7) H_2O has the highest boiling point in the hydride series.
- (8) H_2S has the lowest boiling point among the four hydrides.
- (9) H_2Te has an H-E bond distance smaller than H_2S .
- (10) The H-E-H angle for H_2O is about 90° .
- (11) The bond dissociation enthalpy of H-Te is greater than H-O.
- (12) The numerical data support decreasing thermal stability down the group.
- (13) The numerical dissociation constants support increasing acidity down the group.
- (14) The boiling point of H_2Se exceeds that of H_2Te .
- (15) The H-E bond length decreases from H_2O to H_2Te .

Answer: _____

Q10. Oxides of Group 16 [Count the correct statements.]

- (1) The heavier Group 16 elements form oxides of the EO_2 and EO_3 types.
- (2) Ozone, O_3 , and sulphur dioxide, SO_2 , are gases under ordinary conditions discussed.
- (3) Selenium dioxide, SeO_2 , is a solid.
- (4) The reducing property of the dioxides decreases from SO_2 to TeO_2 .
- (5) SO_2 behaves as a reducing agent.
- (6) TeO_2 is described as an oxidising agent in this trend.
- (7) Sulphur, selenium and tellurium also form EO_3 type oxides: SO_3 , SeO_3 and TeO_3 .
- (8) Both EO_2 and EO_3 oxide types are described as acidic.
- (9) Reducing character of the dioxides increases from SO_2 to TeO_2 .

- (10) TeO_2 is a stronger reducing agent than SO_2 .
- (11) SeO_2 is described as a gas at room temperature.
- (12) SO_3 is described as a basic oxide.
- (13) The +4 oxidation state is represented in EO_2 type oxides such as SO_2 .
- (14) The +6 oxidation state is represented in EO_3 type oxides such as SO_3 .
- (15) Oxygen forms a stable EO_3 oxide of itself in the same sense as SO_3 .

Answer: _____

Q11. Halides of Group 16 [Count the correct statements.]

- (1) Group 16 elements form many halides of the types EX_6 , EX_4 and EX_2 .
- (2) The stability of halides is described as decreasing in the order fluoride > chloride > bromide > iodide.
- (3) Among hexahalides, the hexafluorides are the stable members.
- (4) The hexafluorides are gaseous and have octahedral structures.
- (5) SF_6 is exceptionally stable for steric reasons.
- (6) SF_4 is a gas, SeF_4 a liquid and TeF_4 a solid.
- (7) The tetrafluorides are described with a trigonal-bipyramidal electron arrangement and see-saw molecular geometry.
- (8) One equatorial position in the tetrafluoride arrangement is occupied by a lone pair.
- (9) All Group 16 elements except oxygen form dichlorides and dibromides.
- (10) The well-known monohalides are dimeric, such as S_2F_2 , S_2Cl_2 and Se_2Cl_2 .
- (11) Dimeric halides can undergo disproportionation.
- (12) One disproportionation is $2\text{Se}_2\text{Cl}_2 \rightarrow \text{SeCl}_4 + 3\text{Se}$.
- (13) SF_6 has a tetrahedral structure.
- (14) TeF_4 is a gas.
- (15) Oxygen readily forms stable dichlorides and dibromides of the same type as sulphur.

Answer: _____

Q12. Other sulphur species and paramagnetism [Count the correct statements.]

- (1) Several ring modifications containing about 6-20 sulphur atoms have been synthesised.
- (2) Cyclo- S_6 adopts a chair form.
- (3) At elevated temperatures near 1000 K, S_2 becomes the dominant sulphur species.
- (4) S_2 is paramagnetic like O_2 .
- (5) The paramagnetism of S_2 is associated with two unpaired electrons in antibonding π^* orbitals.
- (6) S_2 is diamagnetic because all electrons are paired.
- (7) Cyclo- S_6 is described as planar hexagonal.
- (8) The atomicity of sulphur changes with temperature.
- (9) Paramagnetic behaviour is specifically associated with sulphur vapour containing S_2 .
- (10) Only S_8 can exist in sulphur vapour at any temperature.
- (11) Rhombic and monoclinic forms differ in crystal packing even though both contain S_8 molecules.
- (12) The α and β forms differ because one contains S_8 and the other S_6 .
- (13) The S_8 structural unit in the common allotropes is non-planar.
- (14) The transition between α and β sulphur changes the molecular formula from S_8 to S_2 .
- (15) The high-temperature S_2 species provides an analogy with O_2 in magnetic behaviour.

Answer: _____

Q13. Advanced oxoacid acidity, oxidation-state and linkage challenge [Count the correct statements.]

- (1) For oxoacids of the same central atom, increasing number of terminal oxo groups generally increases acidity.

- (2) H_2SO_4 is a stronger acid than H_2SO_3 in aqueous solution.
- (3) A peroxo O-O linkage does not require sulphur to have an oxidation state above +6.
- (4) In H_2SO_5 , the oxidation state of sulphur is +6.
- (5) In $\text{H}_2\text{S}_2\text{O}_8$, each sulphur is in the +6 oxidation state.
- (6) In $\text{H}_2\text{S}_2\text{O}_7$, the S-O-S bridge is not a peroxide linkage.
- (7) In a peroxide linkage each oxygen is assigned oxidation state -1 for oxidation-number calculation.
- (8) Sulphur reaches +7 in $\text{H}_2\text{S}_2\text{O}_8$ because the molecule contains eight oxygen atoms.
- (9) Every O-O bond in an oxoacid is assigned oxygen oxidation state -2.
- (10) The presence of O-O versus S-O-S linkages is a key structural distinction among sulphur oxoacids.
- (11) Oxidation-number assignment must be based on bonding/charge rules rather than simply counting oxygen atoms.
- (12) H_2SO_3 and H_2SO_4 contain sulphur in the same oxidation state.
- (13) $\text{H}_2\text{S}_2\text{O}_7$ can be formed by absorption of SO_3 in H_2SO_4 .
- (14) Dilution of $\text{H}_2\text{S}_2\text{O}_7$ produces H_2SO_4 .
- (15) H_2SO_4 contains an O-O peroxide bond.

Answer: _____

Q14. Group 16 reaction-equation mastery [Count the correct statements.]

- (1) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ is a preparation of dioxygen.
- (2) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ is a catalytic preparation of dioxygen.
- (3) $3\text{O}_2 \rightarrow 2\text{O}_3$ represents ozone formation.
- (4) $\text{PbS} + 4\text{O}_3 \rightarrow \text{PbSO}_4 + 4\text{O}_2$ illustrates ozone oxidation.
- (5) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$ is the direct preparation of sulphur dioxide.
- (6) $\text{SO}_3^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O} + \text{SO}_2$ is a laboratory generation of SO_2 .
- (7) $2\text{NaOH} + \text{SO}_2 \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$ occurs with alkali.
- (8) $\text{Na}_2\text{SO}_3 + \text{H}_2\text{O} + \text{SO}_2 \rightarrow 2\text{NaHSO}_3$ occurs with further SO_2 .
- (9) $\text{SO}_2 + \text{Cl}_2 \rightarrow \text{SO}_2\text{Cl}_2$ is catalysed by charcoal.
- (10) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ is the key Contact-process oxidation.
- (11) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ forms oleum.
- (12) $\text{Cu} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$.
- (13) $\text{S} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$.
- (14) $\text{C} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$.
- (15) $\text{SO}_3 + \text{H}_2\text{SO}_4$ directly gives H_2SO_3 .

Answer: _____

Q15. Atomic and physical data - Data mastery II [Count the correct statements.]

- (1) The densities given for O, S, Se and Te are approximately 1.32, 2.06, 4.19 and 6.25 g cm⁻³ respectively under the specified conditions.
- (2) The sulphur density value 2.06 g cm⁻³ corresponds to rhombic sulphur.
- (3) The selenium density value is associated with hexagonal grey selenium.
- (4) Melting points for O, S, Se, Te and Po are approximately 55, 393, 490, 725 and 520 K.
- (5) Boiling points for O, S, Se, Te and Po are approximately 90, 718, 958, 1260 and 1235 K.
- (6) The melting point does not increase monotonically from Te to Po; Po is lower than Te.
- (7) The boiling point of Po is lower than that of Te.
- (8) Oxygen commonly exhibits oxidation states -2 and -1 and positive states +1 and +2 in oxygen fluorides.
- (9) Sulphur, selenium and tellurium show -2, +2, +4 and +6 oxidation states.
- (10) Polonium is represented mainly by +2 and +4 oxidation states in the data table.
- (11) Polonium has a short half-life of about 13.8 days.
- (12) Polonium has a higher boiling point than tellurium in the data table.

- (13) The melting point of oxygen is about 393 K.
 (14) Sulphur is represented only by -2 and +6 oxidation states.
 (15) The physical data show a general rise in melting and boiling points down the group with notable end-of-group deviations.

Answer: _____

Q16. Hydride data - formation enthalpy and dissociation constant [Count the correct statements.]

- (1) Standard enthalpies of formation for H_2O , H_2S , H_2Se and H_2Te are about -286, -20, +73 and +100 kJ mol^{-1} respectively.
 (2) The enthalpy of formation becomes progressively less favourable from H_2O to H_2Te .
 (3) The first dissociation constant of H_2O is about 1.8×10^{-16} at 298 K.
 (4) The first dissociation constant of H_2S is about 1.3×10^{-7} at 298 K.
 (5) The first dissociation constant of H_2Se is about 1.3×10^{-4} at 298 K.
 (6) The first dissociation constant of H_2Te is about 2.3×10^{-3} at 298 K.
 (7) The dissociation constants increase by many orders of magnitude from H_2O to H_2Te .
 (8) The numerical K_a values are consistent with H_2Te being much more acidic than H_2S .
 (9) The H-E bond dissociation enthalpy falls from 463 kJ mol^{-1} for H-O to 238 kJ mol^{-1} for H-Te.
 (10) The first dissociation constant of H_2O is larger than that of H_2Te .
 (11) H_2S has a positive formation enthalpy of about +100 kJ mol^{-1} .
 (12) H_2Te has the strongest H-E bond among the four hydrides.
 (13) Increasing acidity parallels decreasing H-E bond strength down the group.
 (14) The formation-enthalpy data also reflect decreasing thermochemical stability down the group.
 (15) All four hydrides have negative standard enthalpies of formation.

Answer: _____

Q17. ALL USES - Group 16 complete master coverage [Count the correct statements.]

- (1) Dioxygen is essential in normal respiration and supports combustion processes.
 (2) Oxygen is used in oxyacetylene welding.
 (3) Oxygen is used in the manufacture of many metals, particularly steel.
 (4) Oxygen cylinders are used in hospitals, high-altitude flying and mountaineering.
 (5) Liquid oxygen is used with fuels such as hydrazines to provide very high thrust in rockets.
 (6) Ozone is used as a germicide and disinfectant.
 (7) Ozone is used for sterilising water.
 (8) Ozone is used for bleaching oils, ivory, flour and starch.
 (9) Ozone is used as an oxidising agent in the manufacture of potassium permanganate.
 (10) Sulphur dioxide is used in refining petroleum and sugar.
 (11) Sulphur dioxide is used for bleaching wool and silk and also as an anti-chlor, disinfectant and preservative.
 (12) Sulphur dioxide is used to manufacture sulphuric acid, sodium hydrogen sulphite and calcium hydrogen sulphite.
 (13) Liquid sulphur dioxide is used as a solvent for a number of organic and inorganic chemicals.
 (14) Major sulphuric-acid uses include fertilisers, petroleum refining, pigments, paints, dyestuff intermediates, detergents, metallurgical cleaning, storage batteries, nitrocellulose products and laboratory work.
 (15) Ozone, sulphur dioxide and sulphuric acid have no important industrial or laboratory uses.

Answer: _____

Q18. Dioxygen - fine preparation, physical data and exceptions [Count the correct statements.]

- (1) Heating KClO_3 in the presence of MnO_2 is a laboratory method for obtaining dioxygen.
 (2) Oxygen-containing salts such as chlorates, nitrates and permanganates can furnish dioxygen on heating.
 (3) Thermal decomposition of Ag_2O can be represented as $2\text{Ag}_2\text{O} \rightarrow 4\text{Ag} + \text{O}_2$.
 (4) Heating Pb_3O_4 can give PbO and O_2 according to $2\text{Pb}_3\text{O}_4 \rightarrow 6\text{PbO} + \text{O}_2$.
 (5) Thermal decomposition of HgO gives mercury and dioxygen.
 (6) PbO_2 can decompose on heating to PbO and O_2 .
 (7) H_2O_2 readily decomposes to H_2O and O_2 in the presence of catalysts such as finely divided metals or MnO_2 .
 (8) During electrolysis of water, oxygen is liberated at the cathode.
 (9) Industrial dioxygen can be obtained by removing CO_2 and water vapour from air, liquefying the remaining gases and fractionally distilling them.
 (10) Dioxygen is a colourless and odourless gas.
 (11) At 293 K, about 3.08 cm^3 of O_2 dissolves in 100 cm^3 of water.
 (12) Dioxygen liquefies at about 90 K and freezes at about 55 K.
 (13) Oxygen has three stable isotopes: ^{16}O , ^{17}O and ^{18}O .
 (14) Molecular oxygen is diamagnetic despite having an even number of electrons.
 (15) Dioxygen reacts directly with nearly all metals and non-metals except some species such as Au, Pt and some noble gases.

Answer: _____

Q19. Dioxygen - equations, bond enthalpy and complete uses audit [Count the correct statements.]

- (1) Calcium burns in oxygen according to $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$.
 (2) Aluminium reacts with oxygen according to $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$.
 (3) Phosphorus burns in oxygen to form P_4O_{10} .
 (4) Carbon burns in oxygen to form CO_2 .
 (5) Roasting ZnS in oxygen gives ZnO and SO_2 .
 (6) Complete combustion of methane can be written $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$.
 (7) Catalytic oxidation of SO_2 by O_2 in the presence of V_2O_5 gives SO_3 .
 (8) In the presence of CuCl_2 , O_2 can oxidise HCl to Cl_2 and H_2O .
 (9) The $\text{O}=\text{O}$ bond dissociation enthalpy is about 493.4 kJ mol^{-1} .
 (10) External heating is often required to initiate reactions with dioxygen because of the high $\text{O}=\text{O}$ bond dissociation enthalpy.
 (11) Dioxygen is used in oxyacetylene welding.
 (12) Dioxygen is used in manufacture of metals, particularly steel.
 (13) Oxygen cylinders are used in hospitals, high-altitude flying and mountaineering.
 (14) Combustion of fuels such as hydrazines in liquid oxygen can provide tremendous thrust in rockets.
 (15) Platinum reacts rapidly and directly with dioxygen under ordinary conditions.

Answer: _____

Q20. Simple oxides - complete classification and examples [Count the correct statements.]

- (1) A binary compound of oxygen with another element is called an oxide.
 (2) MgO and Al_2O_3 are examples of simple oxides.
 (3) Pb_3O_4 and Fe_3O_4 are examples of mixed oxides.
 (4) An oxide that combines with water to give an acid is termed an acidic oxide.

- (5) SO_2 , Cl_2O_7 , CO_2 and N_2O_5 are examples of acidic oxides.
- (6) SO_2 combines with water to give H_2SO_3 .
- (7) Only non-metal oxides can possess acidic character.
- (8) Mn_2O_7 , CrO_3 and V_2O_5 are metal oxides in high oxidation states that show acidic character.
- (9) An oxide that gives a base with water is called a basic oxide.
- (10) Na_2O , CaO and BaO are examples of basic oxides.
- (11) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ illustrates formation of a base from a basic oxide.
- (12) Metallic oxides are generally basic.
- (13) Al_2O_3 is amphoteric and reacts with acids as well as alkalies.
- (14) CO , NO and N_2O are examples of neutral oxides.
- (15) Neutral oxides necessarily react with water to form strong acids.

Answer: _____

Q21. Ozone - exact preparation, numerical data and thermodynamic stability [Count the correct statements.]

- (1) Ozone is an allotropic form of oxygen.
- (2) Ozone is too reactive to remain for long in the atmosphere at sea level.
- (3) At a height of about 20 km, ozone is formed from atmospheric oxygen in the presence of sunlight.
- (4) The ozone layer protects the earth from excessive ultraviolet radiation.
- (5) Passing a slow dry stream of oxygen through a silent electrical discharge gives about 10% ozone in ozonised oxygen.
- (6) For $3\text{O}_2 \rightarrow 2\text{O}_3$, $\Delta H^\circ(298\text{ K})$ is approximately $+142\text{ kJ mol}^{-1}$.
- (7) Formation of ozone from oxygen is endothermic.
- (8) A silent electrical discharge is used so that excessive heating does not promote decomposition of ozone.
- (9) A battery of ozonisers can be used when ozone concentrations greater than 10% are required.
- (10) Pure ozone has a boiling point close to 101.1 K.
- (11) Pure ozone is a pale blue gas, a dark blue liquid and a violet-black solid.
- (12) Ozone has a characteristic smell.
- (13) Above about 100 parts per million, breathing ozone becomes uncomfortable and may cause headache and nausea.
- (14) Decomposition of ozone to dioxygen is endothermic.
- (15) For conversion of ozone to oxygen, ΔH is negative and ΔS is positive, making the process thermodynamically favourable.

Answer: _____

Q22. Ozone - analysis, atmospheric chemistry, structure and uses [Count the correct statements.]

- (1) $\text{O}_3 \rightarrow \text{O}_2 + \text{O}$ is used to represent the ready liberation of nascent oxygen by ozone.
- (2) Lead sulphide is oxidised by ozone to lead sulphate.
- (3) Ozone oxidises iodide ions to iodine in aqueous medium.
- (4) For quantitative estimation, ozone is passed into excess KI solution buffered with borate buffer at about pH 9.2.
- (5) The iodine liberated in ozone estimation can be titrated with standard sodium thiosulphate solution.
- (6) The iodometric method provides a quantitative estimate of ozone.
- (7) Nitric oxide reacts rapidly with ozone to form NO_2 and O_2 .
- (8) Nitrogen oxides from supersonic jet exhausts have been considered a possible cause of depletion of upper-atmosphere ozone.
- (9) Freons used in aerosol sprays and as refrigerants are another threat to the ozone layer.

- (10) The two O-O bond lengths in ozone are identical and are about 128 pm.
- (11) The O-O-O bond angle in ozone is about 117° .
- (12) Ozone is a linear molecule.
- (13) Ozone is a resonance hybrid of two principal canonical forms.
- (14) Ozone is used as a germicide, disinfectant and for sterilising water.
- (15) Ozone is used for bleaching oils, ivory, flour and starch and as an oxidising agent in manufacture of potassium permanganate.

Answer: _____

Q23. Sulphur allotropes - exact melting points, densities, preparation and structure [Count the correct statements.]

- (1) Rhombic sulphur is the stable form at room temperature.
- (2) Rhombic sulphur transforms into monoclinic sulphur above 369 K.
- (3) Rhombic sulphur is yellow, melts at about 385.8 K and has specific gravity about 2.06.
- (4) Rhombic sulphur crystals can be obtained by evaporating a solution of roll sulphur in CS_2 .
- (5) Rhombic sulphur is insoluble in water, dissolves to some extent in benzene, alcohol and ether, and is readily soluble in CS_2 .
- (6) Monoclinic sulphur melts at about 393 K and has specific gravity about 1.98.
- (7) Monoclinic sulphur can be prepared by melting rhombic sulphur, allowing a crust to form, making holes and pouring out the remaining liquid.
- (8) Monoclinic sulphur forms colourless needle-shaped crystals.
- (9) Monoclinic sulphur is stable above 369 K and changes to rhombic sulphur below 369 K.
- (10) At 369 K both rhombic and monoclinic forms are stable; this is the transition temperature.
- (11) Both rhombic and monoclinic sulphur contain S_8 molecules.
- (12) The S_8 ring in these forms is planar.
- (13) Ring allotropes containing roughly 6-20 sulphur atoms have been synthesised.
- (14) Cyclo- S_6 adopts a chair form.
- (15) At temperatures around 1000 K, S_2 becomes a dominant vapour species and is paramagnetic like O_2 .

Answer: _____

Q24. Sulphur dioxide - fine preparation, physical constants, reactions and uses [Count the correct statements.]

- (1) Burning sulphur in air or oxygen gives SO_2 together with a small amount, about 6-8%, of SO_3 .
- (2) In the laboratory, SO_2 can be generated by treating a sulphite with dilute sulphuric acid.
- (3) Industrial SO_2 is produced as a by-product during roasting of sulphide ores.
- (4) After drying, SO_2 can be liquefied under pressure and stored in steel cylinders.
- (5) SO_2 is a colourless gas with a pungent smell and is highly soluble in water.
- (6) SO_2 can be liquefied at room temperature under a pressure of about two atmospheres.
- (7) The boiling point of SO_2 is about 263 K.
- (8) Passing SO_2 through water gives a solution of sulphurous acid.
- (9) SO_2 reacts with NaOH first to form sodium sulphite; additional SO_2 gives sodium hydrogen sulphite.
- (10) In its reactions with water and alkalies, SO_2 behaves similarly to CO_2 .
- (11) In the presence of charcoal, SO_2 reacts with chlorine to form sulphuryl chloride, SO_2Cl_2 .
- (12) Oxygen oxidises SO_2 to SO_3 in the presence of V_2O_5 .

- (13) Moist SO_2 acts as a reducing agent, converts Fe(III) to Fe(II) and decolourises acidified permanganate.
 (14) The SO_2 molecule is linear.
 (15) Uses of SO_2 include petroleum and sugar refining, bleaching wool and silk, use as anti-chlor, disinfectant and preservative, manufacture of sulphur chemicals, and use of liquid SO_2 as a solvent.

Answer: _____

Q25. Oxoacids of sulphur - complete formula inventory and stability [Count the correct statements.]

- (1) H_2SO_3 is included among important oxoacids of sulphur.
 (2) $\text{H}_2\text{S}_2\text{O}_3$ is an oxoacid of sulphur.
 (3) $\text{H}_2\text{S}_2\text{O}_4$ is included among sulphur oxoacids.
 (4) $\text{H}_2\text{S}_2\text{O}_5$ is included among sulphur oxoacids.
 (5) Acids of the general formula $\text{H}_2\text{S}_x\text{O}_6$ with x from 2 to 5 are included in the sulphur oxoacid family.
 (6) H_2SO_4 is an oxoacid of sulphur.
 (7) $\text{H}_2\text{S}_2\text{O}_7$ is an oxoacid of sulphur.
 (8) H_2SO_5 is an oxoacid of sulphur.
 (9) $\text{H}_2\text{S}_2\text{O}_8$ is an oxoacid of sulphur.
 (10) Every listed oxoacid of sulphur can be isolated indefinitely as a pure stable substance.
 (11) Some sulphur oxoacids are unstable and are known mainly in aqueous solution or through their salts.
 (12) $\text{H}_2\text{S}_2\text{O}_7$ is oleum or pyrosulphuric acid in the sulphuric-acid process.
 (13) H_2SO_5 contains a peroxide linkage.
 (14) $\text{H}_2\text{S}_2\text{O}_8$ contains a peroxide linkage.
 (15) $\text{H}_2\text{S}_2\text{O}_6$ belongs to the $\text{H}_2\text{S}_x\text{O}_6$ series for $x = 2$.

Answer: _____

Q26. Contact process - complete conditions, purification and plant details [Count the correct statements.]

- (1) Sulphuric acid is one of the most important industrial chemicals.
 (2) One step in the Contact process is burning sulphur or sulphide ores in air to generate SO_2 .
 (3) SO_2 is oxidised to SO_3 with oxygen in the presence of V_2O_5 .
 (4) SO_3 is absorbed in concentrated H_2SO_4 to form oleum.
 (5) The SO_2 feed is purified by removing dust and impurities such as arsenic compounds.
 (6) The key equilibrium is $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ with ΔH° approximately $-196.6 \text{ kJ mol}^{-1}$.
 (7) The catalytic oxidation is exothermic, reversible and accompanied by a decrease in gaseous volume.
 (8) Lower temperature and higher pressure favour equilibrium formation of SO_3 .
 (9) Extremely low operating temperature is avoided because the reaction rate would become too slow.
 (10) Industrial operation is carried out at about 2 bar and 720 K.
 (11) Industrial SO_3 is preferably absorbed directly in a large quantity of water to avoid oleum formation.
 (12) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ is the oleum-forming reaction.
 (13) Dilution of oleum with water gives sulphuric acid of the desired concentration.
 (14) Sulphuric acid obtained by the Contact process is about 96-98% pure.
 (15) The plant flow scheme includes dust removal and an arsenic purifier using gelatinous hydrated ferric oxide.

Answer: _____

Q27. Sulphuric acid - exact physical data, ionisation, volatility, dehydration and uses [Count the correct statements.]

- (1) Sulphuric acid is a colourless, dense, oily liquid with specific gravity about 1.84 at 298 K.
 (2) Its freezing point is about 283 K and boiling point about 611 K.

- (3) Dissolving concentrated H_2SO_4 in water evolves a large quantity of heat.
 (4) For safe dilution, water should be added rapidly to concentrated sulphuric acid.
 (5) Concentrated acid should be added slowly to water with constant stirring.
 (6) Important chemical characteristics include low volatility, strong acidity, strong affinity for water and oxidising ability.
 (7) The first ionisation constant is very large, with $K_{a1} > 10$.
 (8) The second ionisation constant is about 1.2×10^{-2} .
 (9) Sulphuric acid forms normal sulphates as well as acid sulphates.
 (10) Because of low volatility, H_2SO_4 can be used to prepare more volatile acids such as HF , HCl and HNO_3 from their salts.
 (11) Concentrated H_2SO_4 can dry many wet gases provided they do not react with the acid.
 (12) Its dehydrating action chars carbohydrates such as sucrose.
 (13) Hot concentrated H_2SO_4 is a moderately strong oxidising agent and is reduced mainly to SO_2 in the cited reactions.
 (14) Major applications include fertilisers, petroleum refining, pigments/paints/dyestuff intermediates, detergents, metallurgical cleaning, storage batteries, nitrocellulose manufacture and laboratory use.
 (15) Phosphoric acid is a stronger oxidising agent than hot concentrated sulphuric acid.

Answer: _____

PART II — GROUP 17

Q28. EXCLUSIVE ORDER QUESTION - periodic properties [Count the correct statements.]

- (1) Atomic radius: $\text{F} < \text{Cl} < \text{Br} < \text{I}$.
 (2) Ionic radius: $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$.
 (3) First ionisation enthalpy: $\text{F} > \text{Cl} > \text{Br} > \text{I}$.
 (4) Electronegativity: $\text{F} > \text{Cl} > \text{Br} > \text{I}$.
 (5) Electron gain enthalpy becomes less negative from Cl to I .
 (6) Magnitude of hydration enthalpy: $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$.
 (7) Melting point: $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$.
 (8) Boiling point: $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$.
 (9) Bond length: $\text{F-F} < \text{Cl-Cl} < \text{Br-Br} < \text{I-I}$.
 (10) Bond dissociation enthalpy follows $\text{Cl-Cl} > \text{Br-Br} > \text{F-F} > \text{I-I}$ for the values.
 (11) Oxidising power in aqueous solution: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.
 (12) Reactivity of elemental halogens decreases down the group.
 (13) Acid strength of hydrogen halides: $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$.
 (14) H-X bond dissociation enthalpy: $\text{H-F} > \text{H-Cl} > \text{H-Br} > \text{H-I}$.
 (15) Electron gain enthalpy follows $\text{F} < \text{Cl} < \text{Br} < \text{I}$ in negativity, with fluorine the most negative.

Answer: _____

Q29. Oxidation states and chemical reactivity [Count the correct statements.]

- (1) All halogens exhibit the -1 oxidation state.
 (2) Chlorine, bromine and iodine also exhibit +1, +3, +5 and +7 oxidation states.
 (3) Higher positive oxidation states occur mainly in compounds with oxygen and fluorine.
 (4) The +4 and +6 oxidation states also occur in oxides and oxoacids of chlorine and bromine.
 (5) Fluorine has no d orbitals in its valence shell in the valence-shell model.
 (6) Fluorine cannot expand its octet in that model.
 (7) Because fluorine is the most electronegative element, it shows only -1 oxidation state in its compounds.
 (8) All halogens are highly reactive.

- (9) They react with metals and non-metals to form halides.
- (10) Elemental halogen reactivity decreases down the group.
- (11) Ready acceptance of an electron explains their oxidising nature.
- (12) F_2 is the strongest oxidising halogen.
- (13) Fluorine commonly shows +7 oxidation state in oxoacids.
- (14) Halogen reactivity increases from fluorine to iodine.
- (15) Iodine is the strongest oxidising halogen.

Answer: _____

Q30. Oxidising power, displacement and reactions with water [Count the correct statements.]

- (1) F_2 oxidises Cl^- , Br^- and I^- .
- (2) Cl_2 oxidises Br^- and I^- .
- (3) Br_2 oxidises I^- .
- (4) I_2 oxidises Br^- spontaneously in the reverse of the normal displacement trend.
- (5) A halogen generally oxidises halide ions of higher atomic number.
- (6) Fluorine oxidises water to oxygen.
- (7) The fluorine-water reaction produces H^+ , F^- and O_2 in ionic form.
- (8) Chlorine and bromine react with water to form the corresponding hydrogen halide and hypohalous acid.
- (9) The reaction of iodine with water is non-spontaneous in this discussion.
- (10) Iodide can be oxidised by oxygen in acidic medium.
- (11) Chlorine cannot oxidise iodide.
- (12) Bromine readily oxidises fluoride to fluorine.
- (13) The decreasing oxidising ability down the group is consistent with the electrode potentials.
- (14) The water reactions provide a direct chemical illustration of relative oxidising power.
- (15) F_2 reacts with water only to form HF; no oxygen is produced.

Answer: _____

Q31. Chlorine - complete uses coverage [Count the correct statements.]

- (1) Chlorine is used for bleaching wood pulp used in paper and rayon manufacture.
- (2) It is used for bleaching cotton and textiles.
- (3) It is used in extraction of gold and platinum.
- (4) It is used in manufacture of dyes.
- (5) It is used in manufacture of drugs.
- (6) It is used in manufacture of organic compounds such as CCl_4 and $CHCl_3$.
- (7) It is used in manufacture of DDT.
- (8) It is used in manufacture of refrigerants.
- (9) It is used for sterilising drinking water.
- (10) It is used in preparation of phosgene, $COCl_2$.
- (11) It is used in preparation of tear gas, CCl_3NO_2 .
- (12) It is used in preparation of mustard gas.
- (13) Chlorine is mainly a reducing agent for food preservation.
- (14) Chlorine is unsuitable for water sterilisation.
- (15) Its industrial importance combines bleaching, disinfection, metallurgy and chemical manufacture.

Answer: _____

Q32. Hydrogen chloride - uses [Count the correct statements.]

- (1) HCl is used in the manufacture of chlorine.
- (2) It is used in manufacture of ammonium chloride.
- (3) It is used in manufacture of glucose from corn starch.
- (4) It is used for extracting glue from bones.
- (5) It is used for purifying bone black.
- (6) It has medicinal uses.
- (7) It is used as a laboratory reagent.
- (8) HCl is as the principal raw material for manufacture of ozone.
- (9) HCl is stated to be unusable in laboratory chemistry because it is too weak.

- (10) Its use in chlorine manufacture links hydrogen chloride directly with halogen chemistry.
- (11) Its reaction with weaker-acid salts makes it a useful general laboratory acid.
- (12) HCl is a bleaching agent for wool and silk.
- (13) Hydrochloric acid is an important reagent in both inorganic preparation and analysis.
- (14) HCl is used as the oxidising component of the Contact process.
- (15) The uses include both industrial and laboratory applications.

Answer: _____

Q33. Advanced oxoacid acidity and oxidation-state challenge [Count the correct statements.]

- (1) For chlorine oxoacids, acid strength increases in the order $HClO < HClO_2 < HClO_3 < HClO_4$.
- (2) For oxoacids with the same central halogen, increasing number of terminal oxo groups strengthens the acid.
- (3) The increase in acid strength is associated with stronger electron withdrawal and greater stabilisation of the conjugate base.
- (4) $HClO_4$ contains chlorine in the highest common +7 oxidation state.
- (5) $HClO$ contains chlorine in the +1 oxidation state.
- (6) Within analogous HOX acids, greater electronegativity of X generally strengthens the acid.
- (7) $HOCl$ is therefore stronger than $HOBr$, which is stronger than HOI .
- (8) $HClO$ is stronger than $HClO_4$.
- (9) Increasing the oxidation state of chlorine from +1 to +7 weakens the corresponding oxoacid.
- (10) All halogen oxoacids contain fluorine in a positive oxidation state.
- (11) Oxidation state and number of oxo groups are distinct but related structural variables in oxoacid acidity.
- (12) The O-H bond is the ionisable acidic bond in the usual structural formulas $HO-X(O)_n$.
- (13) $HClO_4$ has chlorine in the -1 oxidation state.
- (14) The oxoacid formulas are consistent with one X-OH bond in each structure.
- (15) HOX proves that fluorine commonly exhibits +1 oxidation state.

Answer: _____

Q34. Interhalogens - reactivity, hydrolysis and uses [Count the correct statements.]

- (1) Most interhalogen compounds are volatile solids or liquids at 298 K, with ClF being a gas.
- (2) Their physical properties are generally intermediate between those of the constituent halogens.
- (3) Their melting and boiling points can be somewhat higher than simple expectation.
- (4) Interhalogens are generally more reactive than the halogens, except fluorine.
- (5) The enhanced reactivity is associated with weaker X-X' bonds than X-X bonds, except for the F-F anomaly.
- (6) Interhalogens undergo hydrolysis.
- (7) Hydrolysis gives halide ion derived from the smaller halogen.
- (8) The larger halogen appears in an oxyanion whose oxidation level depends on XX' , XX'_3 , XX'_5 or XX'_7 .
- (9) XX' hydrolysis gives a hypohalite-type product of the larger halogen.
- (10) XX'_3 , XX'_5 and XX'_7 correspond to halite, halate and perhalate-type oxyanions respectively.
- (11) XX'_3 compounds have bent T shapes.
- (12) XX'_5 compounds are square pyramidal.
- (13) IF_7 is pentagonal bipyramidal.
- (14) ClF_3 and BrF_3 are used as fluorinating agents in production of UF_6 for uranium enrichment.
- (15) $U + 3ClF_3 \rightarrow UF_6 + 3ClF$ is a fluorination reaction.

Answer: _____

Q35. BrF₃ VSEPR and geometry [Count the correct statements.]

- (1) Bromine has seven valence electrons in BrF₃.
- (2) Three electron pairs form Br-F bonds.
- (3) Four electrons remain on bromine as two lone pairs.
- (4) BrF₃ therefore has three bond pairs and two lone pairs around the central atom.
- (5) Five electron pairs are arranged on a trigonal-bipyramidal electron-pair framework.
- (6) The two lone pairs occupy equatorial positions.
- (7) Equatorial lone pairs minimise lone-pair-lone-pair and lone-pair-bond-pair repulsions.
- (8) The axial fluorine atoms are bent slightly toward the equatorial fluorine.
- (9) The molecular shape is a slightly bent T.
- (10) Both lone pairs occupy axial positions.
- (11) BrF₃ has only one lone pair on bromine.
- (12) The electron-pair geometry is tetrahedral.
- (13) Lone-pair repulsions are greater than bond-pair-bond-pair repulsions in the VSEPR explanation.
- (14) The observed molecular geometry differs from the full five-position electron-domain geometry.
- (15) BrF₃ is square planar according to VSEPR.

Answer: _____

Q36. Group 17 reaction-equation mastery [Count the correct statements.]

- (1) $F_2 + 2Cl^- \rightarrow 2F^- + Cl_2$ is consistent with fluorine's oxidising power.
- (2) $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$.
- (3) $Br_2 + 2I^- \rightarrow 2Br^- + I_2$.
- (4) $2F_2 + 2H_2O \rightarrow 4H^+ + 4F^- + O_2$.
- (5) $Cl_2 + H_2O \rightleftharpoons HCl + HOCl$ represents chlorine hydrolysis.
- (6) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$ prepares chlorine.
- (7) $4HCl + O_2 \rightarrow 2Cl_2 + 2H_2O$ is the Deacon reaction.
- (8) $8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$ occurs with excess ammonia.
- (9) $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$ occurs with hot concentrated alkali.
- (10) $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ occurs photochemically.
- (11) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$ is the lower-temperature HCl preparation step.
- (12) $NH_3 + HCl \rightarrow NH_4Cl$ gives white fumes.
- (13) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$.
- (14) $I_2 + Cl_2 \rightarrow 2ICl$ is an interhalogen preparation.
- (15) $Cl_2 + 2F^- \rightarrow 2Cl^- + F_2$ occurs spontaneously in aqueous solution.

Answer: _____

Q37. Group 17 cumulative exceptions and high-yield traps [Count the correct statements.]

- (1) Cl has more negative electron gain enthalpy than F.
- (2) F₂ has lower bond dissociation enthalpy than Cl₂.
- (3) HF has a much higher boiling point than HCl because of hydrogen bonding.
- (4) F₂ is nevertheless the strongest oxidising halogen.
- (5) Fluorine exhibits only -1 oxidation state in its compounds.
- (6) Iodine reaction with water is non-spontaneous in this comparison.
- (7) Bromine oxides are the least stable among the compared Cl, Br and I oxides.
- (8) Higher halogen oxides tend to be more stable than lower oxides.
- (9) Chlorine gives hypochlorite in cold dilute alkali but chlorate in hot concentrated alkali.
- (10) Excess ammonia and excess chlorine give different nitrogen-containing products.
- (11) Bleaching by chlorine is oxidative and permanent.
- (12) HCl is dried using concentrated H₂SO₄.
- (13) Fluorine forms only one oxoacid, HOF.
- (14) Interhalogens are generally more reactive than the parent halogens except F₂.

- (15) All down-group trends in Group 17 are perfectly regular with no important fluorine anomalies.

Answer: _____

Q38. All colours, physical descriptions and visual-identification facts [Count the correct statements.]

- (1) F₂ is yellow and Cl₂ is greenish yellow.
- (2) Br₂ is red.
- (3) I₂ is violet.
- (4) Chlorine is a greenish-yellow gas and can be liquefied to a greenish-yellow liquid.
- (5) Hydrogen chloride is a colourless gas and forms a colourless liquid on liquefaction.
- (6) Solid HCl is described as white crystalline.
- (7) ClF is colourless.
- (8) BrF is pale brown.
- (9) α-ICl is ruby red and β-ICl is brown red.
- (10) IBr is black.
- (11) ClF₃ is colourless, BrF₃ yellow green and IF₃ yellow.
- (12) ICl₃ is orange.
- (13) IF₅ and IF₇ are colourless.
- (14) BrF₃ is a violet gas.
- (15) IBr is a colourless liquid.

Answer: _____

Q39. All uses - halogen and interhalogen master coverage [Count the correct statements.]

- (1) ClO₂ is used for bleaching paper pulp and textiles.
- (2) ClO₂ is used in water treatment.
- (3) I₂O₅ is used in estimation of carbon monoxide.
- (4) Chlorine is used for bleaching wood pulp, cotton and textiles.
- (5) Chlorine is used in extraction of gold and platinum.
- (6) Chlorine is used in manufacture of dyes and drugs.
- (7) Chlorine is used in manufacture of CCl₄, CHCl₃, DDT and refrigerants.
- (8) Chlorine is used to sterilise drinking water.
- (9) Chlorine is used to prepare phosgene, tear gas and mustard gas.
- (10) HCl is used to manufacture chlorine and NH₄Cl.
- (11) HCl is used to manufacture glucose from corn starch.
- (12) HCl is used in extraction of glue from bones and purification of bone black.
- (13) HCl is used in medicine and as a laboratory reagent.
- (14) Interhalogens can be used as non-aqueous solvents and as fluorinating agents.
- (15) ClF₃ and BrF₃ are used for UF₆ production in uranium enrichment.

Answer: _____

Q40. Atomic and physical data - Data mastery II [Count the correct statements.]

- (1) Melting points of F₂, Cl₂, Br₂ and I₂ are approximately 54.4, 172.0, 265.8 and 386.6 K.
- (2) Boiling points of F₂, Cl₂, Br₂ and I₂ are approximately 84.9, 239.0, 332.5 and 458.2 K.
- (3) The density values given are about 1.5, 1.66, 3.19 and 4.94 g cm⁻³ for F₂, Cl₂, Br₂ and I₂ under the specified conditions.
- (4) Hydration enthalpy magnitudes of F⁻, Cl⁻, Br⁻ and I⁻ are about 515, 381, 347 and 305 kJ mol⁻¹.
- (5) Standard reduction potentials E° are about 2.87, 1.36, 1.09 and 0.54 V for F₂/F⁻, Cl₂/Cl⁻, Br₂/Br⁻ and I₂/I⁻.
- (6) F₂ has the lowest melting and boiling points among F₂, Cl₂, Br₂ and I₂.
- (7) I₂ has the highest melting and boiling points among these four.
- (8) The standard reduction potential falls sharply from fluorine to iodine.
- (9) The hydration enthalpy magnitude of F⁻ greatly exceeds that of I⁻.
- (10) The boiling point of Br₂ is lower than that of Cl₂.
- (11) The standard reduction potential of I₂/I⁻ exceeds that of F₂/F⁻.
- (12) Seaweeds can contain up to about 0.5% iodine.

- (13) Chile saltpetre can contain up to about 0.2% sodium iodate.
 (14) Sea water is described as containing about 25% NaCl by mass.
 (15) The physical data quantitatively support the increase of dispersion forces and phase-change temperatures down the group.

Answer: _____

Q41. Interhalogen table - complete fine-detail audit [Count the correct statements.]

- (1) IF is very unstable and has been detected spectroscopically.
 (2) BrCl is an interhalogen compound of the XX' type.
 (3) ICl exists in two solid forms with different colours.
 (4) ICl₃ can dimerise as a chlorine-bridged species, I₂Cl₆.
 (5) ClF₃, BrF₃, IF₃ and ICl₃ belong to the XX'₃ class.
 (6) IF₅, BrF₅ and ClF₅ belong to the XX'₅ class.
 (7) IF₇ is the representative XX'₇ compound.
 (8) XX'₃ compounds are assigned bent T-shaped geometries.
 (9) XX'₅ compounds are square pyramidal.
 (10) IF₇ is pentagonal bipyramidal.
 (11) The interhalogens are covalent and diamagnetic.
 (12) Their physical properties are generally intermediate between those of the constituent halogens.
 (13) Their melting and boiling points may be somewhat higher than simple interpolation would suggest.
 (14) All interhalogens are ionic solids at 298 K.
 (15) IF₇ belongs to the XX'₅ class.

Answer: _____

Q42. Hydrogen chloride - quantitative and analytical fine details [Count the correct statements.]

- (1) The first laboratory step for HCl preparation is carried out near 420 K.
 (2) The second NaHSO₄ + NaCl step is carried out near 823 K.
 (3) HCl has a boiling point near 189 K and freezing point near 159 K.
 (4) The dissociation constant quoted for HCl in water is of the order of 10⁷.
 (5) The very large dissociation constant is consistent with strong-acid behaviour in water.
 (6) Concentrated H₂SO₄ is suitable for drying HCl gas.
 (7) HCl gives dense white fumes of NH₄Cl with NH₃.
 (8) Three volumes of concentrated HCl and one volume of concentrated HNO₃ form aqua regia.
 (9) Aqua regia dissolves gold and platinum by combined acidic, oxidising and complexing action.
 (10) Carbonates and hydrogencarbonates release CO₂ with HCl.
 (11) Sulphites release SO₂ with HCl.
 (12) Finely powdered iron gives FeCl₂ rather than FeCl₃ with HCl.
 (13) HCl is dried by passing it through concentrated NaOH.
 (14) The HCl/HNO₃ volume ratio in aqua regia is 1:3.
 (15) The combination of high water solubility and extensive ionisation explains the strong acidic action of aqueous HCl.

Answer: _____

Q43. ALL REACTION CONDITIONS AND OBSERVATIONS - Group 17 master audit [Count the correct statements.]

- (1) MnO₂ is heated with concentrated HCl to prepare chlorine in the laboratory.
 (2) Deacon's process oxidises HCl with atmospheric oxygen over CuCl₂ at about 723 K.
 (3) Chlorine is liberated at the anode during electrolysis of concentrated NaCl solution.
 (4) With excess NH₃, chlorine gives N₂ and NH₄Cl; with excess Cl₂, explosive NCl₃ is formed.

- (5) Cold dilute NaOH converts chlorine into chloride and hypochlorite.
 (6) Hot concentrated NaOH converts chlorine into chloride and chlorate.
 (7) Dry slaked lime reacts with chlorine to give bleaching powder.
 (8) Methane undergoes chlorination under ultraviolet light, whereas ethene gives an addition product with chlorine at room temperature.
 (9) The first NaCl-concentrated H₂SO₄ step for HCl preparation is carried out near 420 K.
 (10) The NaHSO₄ + NaCl step for HCl preparation is carried out near 823 K.
 (11) Hydrogen chloride gas is dried by passing it through concentrated H₂SO₄.
 (12) Aqua regia is prepared by mixing three volumes of concentrated HCl with one volume of concentrated HNO₃.
 (13) Equal volumes of Cl₂ and F₂ at about 437 K give ClF, while excess F₂ at about 573 K gives ClF₃.
 (14) Br₂ with 3F₂ gives BrF₃ under controlled conditions, while excess fluorine can give BrF₅.
 (15) Cold dilute alkali and hot concentrated alkali give exactly the same chlorine disproportionation product mixture.

Answer: _____

Q44. Occurrence of halogens - complete natural- and abundance audit [Count the correct statements.]

- (1) Fluorine, chlorine, bromine, iodine and astatine are the elements of Group 17.
 (2) The name halogen comes from Greek roots meaning salt producer.
 (3) Halogens are highly reactive non-metallic elements.
 (4) Astatine is radioactive.
 (5) Fluorine and chlorine are fairly abundant, whereas bromine and iodine are less abundant.
 (6) Fluorine occurs mainly as insoluble fluorides such as fluorspar, CaF₂.
 (7) Cryolite has the formula Na₃AlF₆.
 (8) Fluoroapatite can be represented as 3Ca₃(PO₄)₂·CaF₂.
 (9) Small quantities of fluorine occur in soil, river water, plants, and in animal bones and teeth.
 (10) Sea water contains chlorides, bromides and iodides of sodium, potassium, magnesium and calcium.
 (11) Sea water is mainly sodium chloride solution, about 2.5% by mass.
 (12) Deposits of dried-up seas contain substances such as NaCl and carnallite, KCl·MgCl₂·6H₂O.
 (13) Certain seaweeds may contain up to about 0.5% iodine.
 (14) Chile saltpetre can contain up to about 0.2% sodium iodate.
 (15) Iodine is more abundant than chlorine in sea water.

Answer: _____

Q45. Halogen table - exact atomic, thermochemical and physical data [Count the correct statements.]

- (1) Atomic numbers of F, Cl, Br, I and At are 9, 17, 35, 53 and 85.
 (2) Approximate atomic masses are 19.00, 35.45, 79.90, 126.90 and 210 g mol⁻¹.
 (3) Covalent radii of F, Cl, Br and I are approximately 64, 99, 114 and 133 pm.
 (4) X⁻ ionic radii of F⁻, Cl⁻, Br⁻ and I⁻ are approximately 133, 184, 196 and 220 pm.
 (5) First ionisation enthalpies of F, Cl, Br and I are approximately 1680, 1256, 1142 and 1008 kJ mol⁻¹.
 (6) Electron gain enthalpies of F, Cl, Br and I are about -333, -349, -325 and -296 kJ mol⁻¹.
 (7) Electronegativities are approximately 4.0, 3.2, 3.0, 2.7 and 2.2 for F, Cl, Br, I and At.
 (8) Magnitudes of hydration enthalpy for F⁻, Cl⁻, Br⁻ and I⁻ decrease roughly as 515, 381, 347 and 305 kJ mol⁻¹.

- (9) Melting points of F_2 , Cl_2 , Br_2 and I_2 are about 54.4, 172.0, 265.8 and 386.6 K.
 (10) Boiling points of F_2 , Cl_2 , Br_2 and I_2 are about 84.9, 239.0, 332.5 and 458.2 K.
 (11) X-X bond distances for F_2 , Cl_2 , Br_2 and I_2 are about 143, 199, 228 and 266 pm.
 (12) Bond dissociation enthalpies are about 158.8, 242.6, 192.8 and 151.1 kJ mol⁻¹ for F_2 , Cl_2 , Br_2 and I_2 .
 (13) Standard reduction potentials for F_2/F^- , Cl_2/Cl^- , Br_2/Br^- and I_2/I^- are approximately 2.87, 1.36, 1.09 and 0.54 V.
 (14) Electron gain enthalpy of fluorine is more negative than that of chlorine.
 (15) Among F_2 , Cl_2 , Br_2 and I_2 , the Cl-Cl bond has the smallest bond dissociation enthalpy.

Answer: _____

Q46. Halogen physical properties - colours, states, solubility and bond anomaly [Count the correct statements.]

- (1) All Group 17 elements have seven electrons in the outermost shell with general configuration ns^2np^5 .
 (2) Halogens have the smallest atomic radii in their respective periods because of high effective nuclear charge.
 (3) Atomic and ionic radii increase from fluorine to iodine.
 (4) Ionisation enthalpy is high for halogens and decreases down the group.
 (5) Halogens have very negative electron gain enthalpies in their respective periods.
 (6) Electron gain enthalpy generally becomes less negative down the group, but fluorine is less negative than chlorine.
 (7) The relatively small 2p shell of fluorine leads to strong interelectronic repulsion for an incoming electron.
 (8) Electronegativity decreases down Group 17 and fluorine is the most electronegative element.
 (9) F_2 and Cl_2 are gases, Br_2 is a liquid and I_2 is a solid under ordinary conditions.
 (10) Melting and boiling points increase steadily down the group.
 (11) All halogens are coloured because visible radiation can excite outer electrons.
 (12) F_2 is yellow, Cl_2 greenish-yellow, Br_2 red and I_2 violet.
 (13) Fluorine and chlorine react with water, whereas bromine and iodine are only sparingly soluble in water.
 (14) Bromine and iodine dissolve in organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions.
 (15) The F-F bond is stronger than the Cl-Cl bond.

Answer: _____

Q47. Fluorine anomaly and hydrogen halides - complete numerical trend audit [Count the correct statements.]

- (1) Anomalous behaviour of fluorine is associated with small size, highest electronegativity, low F-F bond dissociation enthalpy and absence of valence d orbitals.
 (2) Most reactions of fluorine are exothermic because strong bonds are formed with other elements.
 (3) Fluorine forms only one oxoacid in the cited series.
 (4) HF is a liquid with boiling point about 293 K because of strong hydrogen bonding.
 (5) The other common hydrogen halides are gases under ordinary conditions.
 (6) Affinity of halogens for hydrogen decreases from fluorine to iodine.
 (7) Aqueous acid strength follows $HF < HCl < HBr < HI$.
 (8) Thermal stability follows $HF > HCl > HBr > HI$.
 (9) H-X bond lengths for HF, HCl, HBr and HI are about 91.7, 127.4, 141.4 and 160.9 pm.
 (10) H-X bond dissociation enthalpies are about 574, 432, 363 and 295 kJ mol⁻¹ respectively.

- (11) Approximate pK_a values listed are 3.2, -7.0, -9.5 and -10.0 for HF, HCl, HBr and HI.
 (12) HF has a melting point around 190 K and boiling point around 293 K.
 (13) HCl has a boiling point near 189 K.
 (14) HI is a stronger aqueous acid than HBr.
 (15) HF has the weakest H-X bond among the hydrogen halides.

Answer: _____

Q48. Halogen oxides - complete stability, oxidising action and special uses [Count the correct statements.]

- (1) Fluorine forms OF_2 and O_2F_2 .
 (2) Of these two oxygen fluorides, only OF_2 is thermally stable at 298 K.
 (3) OF_2 and O_2F_2 are described as oxygen fluorides because fluorine is more electronegative than oxygen.
 (4) Both OF_2 and O_2F_2 are strong fluorinating agents.
 (5) O_2F_2 can oxidise plutonium to PuF_6 , a reaction used in removal of plutonium from spent nuclear fuel.
 (6) Chlorine, bromine and iodine form oxides in which halogen oxidation states range from +1 to +7.
 (7) The general stability order quoted for halogen oxides is $I > Cl > Br$.
 (8) Higher oxides of halogens tend to be more stable than lower oxides.
 (9) Cl_2O , ClO_2 , Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and can be explosive.
 (10) ClO_2 is used for bleaching paper pulp and textiles and in water treatment.
 (11) Br_2O , BrO_2 and BrO_3 are among the least stable halogen oxides, exist only at low temperatures and are strong oxidants.
 (12) I_2O_4 , I_2O_5 and I_2O_7 are insoluble solids that decompose on heating.
 (13) I_2O_5 is used in estimation of carbon monoxide because it is a good oxidising agent.
 (14) Bromine oxides are the most stable oxides among Cl, Br and I.
 (15) OF_2 is more appropriately described as an oxygen fluoride than as an oxide of fluorine.

Answer: _____

Q49. Metal halides and interhalogen principles - covalency and composition [Count the correct statements.]

- (1) Bromine reacts with magnesium to form $MgBr_2$.
 (2) For a monovalent metal M, ionic character decreases in the order $MF > MCl > MBr > MI$.
 (3) When a metal has more than one oxidation state, the halide in the higher oxidation state is generally more covalent.
 (4) $SnCl_4$ is more covalent than $SnCl_2$.
 (5) $PbCl_4$ is more covalent than $PbCl_2$.
 (6) $SbCl_5$ is more covalent than $SbCl_3$.
 (7) UF_6 is more covalent than UF_4 .
 (8) Interhalogen compounds occur in types XX' , XX'_3 , XX'_5 and XX'_7 .
 (9) In these formulas X is generally the larger halogen and X' the smaller halogen.
 (10) X is more electropositive than X' in the usual interhalogen notation.
 (11) As the radius ratio X/X' becomes larger, the maximum number of atoms that can be accommodated can increase.
 (12) IF_7 has the maximum atom count among the common interhalogen types listed.
 (13) Interhalogen compounds are predominantly ionic salts.
 (14) Interhalogens listed are covalent molecules and diamagnetic.
 (15) Interhalogen compounds are generally more reactive than the constituent halogens except fluorine.

Answer: _____

Q50. Chlorine - discovery, laboratory preparation, manufacture and physical data [Count the correct statements.]

- (1) Chlorine was discovered by Scheele in 1774 by the action of HCl on MnO₂.
- (2) Davy established its elementary nature in 1810 and the name chlorine reflects its greenish-yellow colour.
- (3) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ is a laboratory preparation reaction.
- (4) A mixture of common salt and concentrated H₂SO₄ can be used in place of ready-made HCl in the MnO₂ method.
- (5) An overall preparation equation is $4\text{NaCl} + \text{MnO}_2 + 4\text{H}_2\text{SO}_4 \rightarrow \text{MnCl}_2 + 4\text{NaHSO}_4 + 2\text{H}_2\text{O} + \text{Cl}_2$.
- (6) KMnO₄ oxidises HCl according to $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 8\text{H}_2\text{O} + 5\text{Cl}_2$.
- (7) In the Deacon process HCl is oxidised by atmospheric oxygen using CuCl₂ catalyst at about 723 K.
- (8) Electrolysis of concentrated NaCl solution liberates chlorine at the anode.
- (9) Chlorine is also obtained as a by-product in several chemical industries.
- (10) Chlorine is a greenish-yellow gas with pungent and suffocating odour.
- (11) Chlorine is heavier than air.
- (12) It can be liquefied to a greenish-yellow liquid boiling at about 239 K.
- (13) Chlorine is soluble in water.
- (14) Elemental chlorine is colourless.
- (15) In brine electrolysis, chlorine is liberated at the cathode.

Answer: _____

Q51. Chlorine - element, ammonia, alkali and bleaching-powder reaction set [Count the correct statements.]

- (1) $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$ is a reaction of chlorine with aluminium.
- (2) $\text{P}_4 + 6\text{Cl}_2 \rightarrow 4\text{PCl}_3$ is a cited reaction of chlorine with phosphorus.
- (3) $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ represents formation of sodium chloride.
- (4) $\text{S}_8 + 4\text{Cl}_2 \rightarrow 4\text{S}_2\text{Cl}_2$ is a cited sulphur-chlorine reaction.
- (5) $2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$ shows formation of ferric chloride.
- (6) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ illustrates the strong affinity of chlorine for hydrogen.
- (7) $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$ shows oxidation of hydrogen sulphide by chlorine.
- (8) $\text{C}_{10}\text{H}_{16} + 8\text{Cl}_2 \rightarrow 16\text{HCl} + 10\text{C}$ is a cited reaction of chlorine with a hydrogen-containing compound.
- (9) With excess NH₃, $8\text{NH}_3 + 3\text{Cl}_2 \rightarrow 6\text{NH}_4\text{Cl} + \text{N}_2$.
- (10) With excess chlorine, $\text{NH}_3 + 3\text{Cl}_2 \rightarrow \text{NCl}_3 + 3\text{HCl}$, and NCl₃ is explosive.
- (11) Cold dilute NaOH converts chlorine into chloride and hypochlorite.
- (12) Hot concentrated NaOH converts chlorine into chloride and chlorate.
- (13) With dry slaked lime, chlorine gives a mixture containing hypochlorite and chloride species used as bleaching powder.
- (14) Bleaching powder is represented as Ca(OCl)₂·CaCl₂·Ca(OH)₂·2H₂O.
- (15) Chlorine reacts with ethene only by substitution and does not add across the double bond.

Answer: _____

Q52. Chlorine water - hydrocarbon reactions, oxidation equations and complete uses [Count the correct statements.]

- (1) Under UV light, $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ is a substitution reaction.
- (2) At room temperature, ethene adds chlorine to give 1,2-dichloroethane.
- (3) Chlorine water loses its yellow colour on standing because HCl and HOCl are formed.

- (4) HOCl can furnish active oxygen responsible for oxidising and bleaching action.
- (5) Chlorine oxidises Fe(II) to Fe(III).
- (6) Chlorine oxidises sulphite to sulphate.
- (7) $\text{SO}_2 + 2\text{H}_2\text{O} + \text{Cl}_2 \rightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$ is a cited oxidation reaction.
- (8) $\text{I}_2 + 6\text{H}_2\text{O} + 5\text{Cl}_2 \rightarrow 2\text{HIO}_3 + 10\text{HCl}$ shows oxidation of iodine to iodic acid.
- (9) Bleaching action of chlorine is an oxidation process.
- (10) Bleaching by chlorine is permanent.
- (11) Chlorine is used in bleaching wood pulp, cotton and textiles.
- (12) Chlorine is used in extraction of gold and platinum.
- (13) It is used in manufacture of dyes, drugs and compounds such as CCl₄, CHCl₃, DDT and refrigerants.
- (14) It is used for sterilising drinking water.
- (15) Chlorine is used in preparation of poisonous gases such as phosgene, tear gas and mustard gas.

Answer: _____

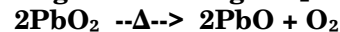
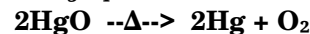
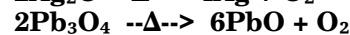
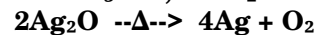
Q53. Halogen oxoacids and interhalogen preparation - exact formula and reaction audit [Count the correct statements.]

- (1) Because of its very high electronegativity and small size, fluorine forms only one oxoacid, HOF.
- (2) Many oxoacids of the other halogens cannot be isolated pure and are stable mainly in aqueous solution or as salts.
- (3) HOF, HOCl, HOBr and HOI are the +1 hypohalous acids listed.
- (4) HClO₂ is the +3 chlorous acid listed in the series.
- (5) HClO₃, HBrO₃ and HIO₃ correspond to the +5 halic acids.
- (6) HClO₄, HBrO₄ and HIO₄ correspond to the +7 perhalic acids in the table.
- (7) In the oxoacid structures the halogen is central and is bonded through an X-OH group; additional X=O bonds occur in higher members.
- (8) Equimolar I₂ and Cl₂ combine to form ICl.
- (9) With excess fluorine at about 573 K, chlorine forms ClF₃.
- (10) Equal volumes of Cl₂ and F₂ at about 437 K form ClF.
- (11) I₂ with excess chlorine can form ICl₃.
- (12) Br₂ with three equivalents of F₂ can form BrF₃.
- (13) Br₂ with five equivalents of F₂ can form BrF₅.
- (14) Interhalogen compounds are covalent and diamagnetic; most are volatile solids or liquids, while ClF is a gas.
- (15) On hydrolysis, the smaller halogen gives halide while the larger halogen gives an oxyanion whose oxidation level depends on XX', XX'₃, XX'₅ or XX'₇ stoichiometry.

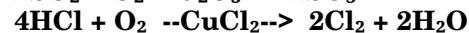
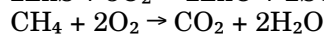
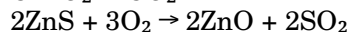
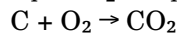
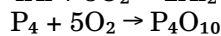
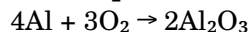
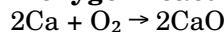
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REACTION BANK — GROUP 16

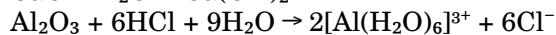
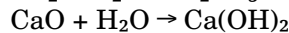
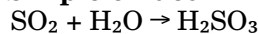
Dioxygen preparation



Dioxygen reactions and catalytic oxidations

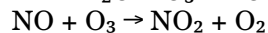
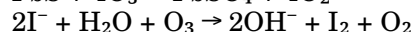
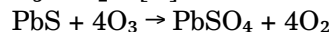
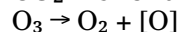


Simple oxides

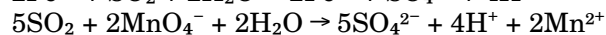
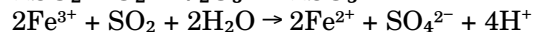
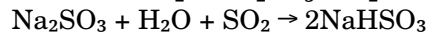
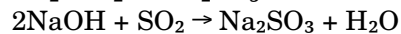
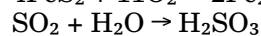
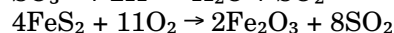
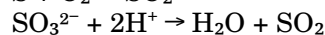
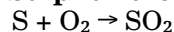


Ozone

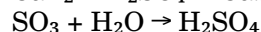
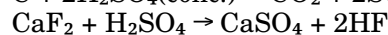
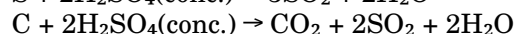
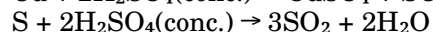
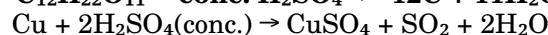
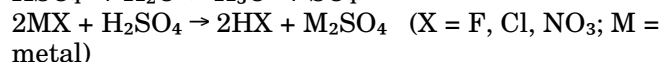
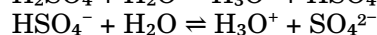
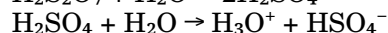
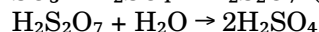
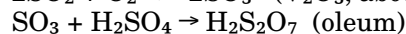
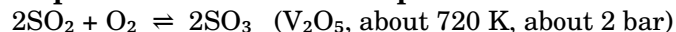
3O₂ --silent electrical discharge--> 2O₃



Sulphur dioxide

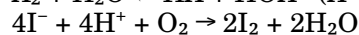
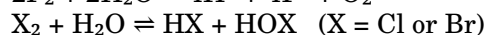
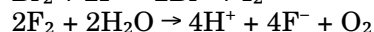
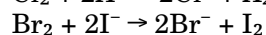
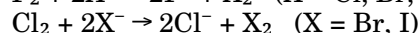
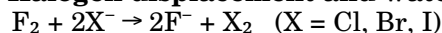


Sulphuric acid and Contact process

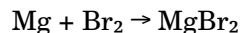


REACTION BANK — GROUP 17

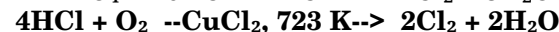
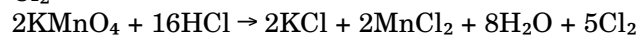
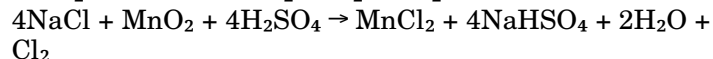
Halogen displacement and water reactions



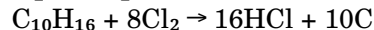
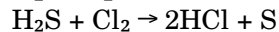
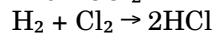
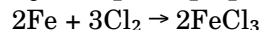
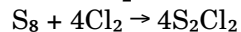
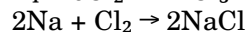
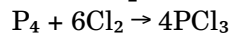
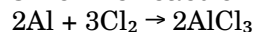
Metal halide formation



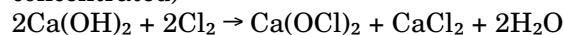
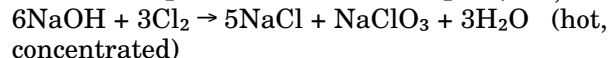
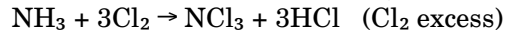
Chlorine preparation and manufacture



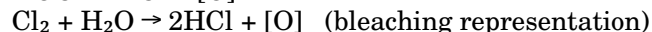
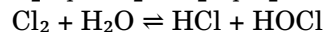
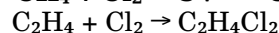
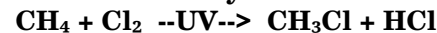
Chlorine reactions with elements and compounds



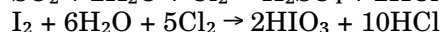
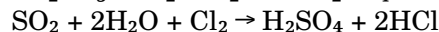
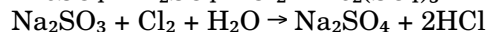
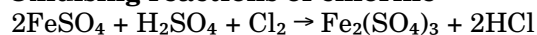
Chlorine with ammonia, alkali and lime



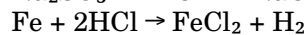
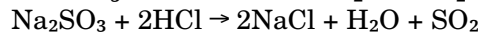
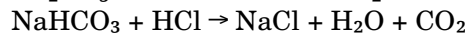
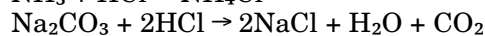
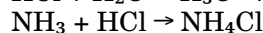
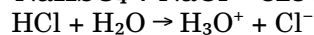
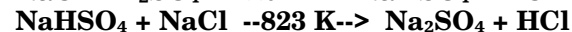
Chlorine with hydrocarbons and water



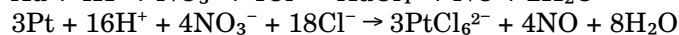
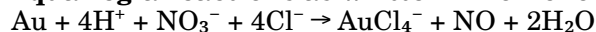
Oxidising reactions of chlorine



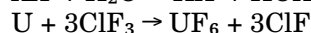
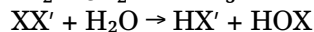
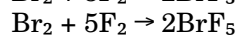
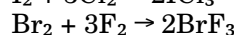
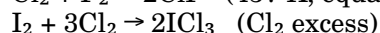
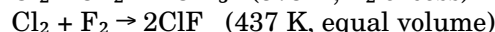
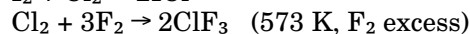
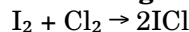
Hydrogen chloride



Aqua regia reactions as written in ionic form



Interhalogen preparation and use



ANSWER KEY

PART I — GROUP 16

Q1: 14 Q2: 10 Q3: 14 Q4: 10 Q5: 10 Q6: 11 Q7: 10

Q8: 11 Q9: 10 Q10: 10 Q11: 12 Q12: 10

Q13: 11 Q14: 14 Q15: 12 Q16: 11 Q17: 14 Q18: 13

Q19: 14 Q20: 13 Q21: 14 Q22: 14 Q23: 14 Q24: 14

Q25: 14 Q26: 14 Q27: 13

PART II — GROUP 17

Q28: 14 Q29: 12 Q30: 11 Q31: 13 Q32: 11 Q33: 10

Q34: 15 Q35: 11 Q36: 14 Q37: 14 Q38: 13 Q39: 15

Q40: 12 Q41: 13 Q42: 13 Q43: 14 Q44: 14 Q45: 13

Q46: 14 Q47: 14 Q48: 14 Q49: 14 Q50: 13 Q51: 14

Q52: 15 Q53: 15